

Predicting Reasoning-Model Costs from Shared Prefill Activations

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Abstract

Routers choose a model per query by trading predicted correctness against cost, but most price each model at a constant, although a reasoning model’s output length varies by an order of magnitude across queries. We predict route-specific output length from the prefill activations of a frozen 4B encoder that already supplies the success predictions, using one linear readout per route. On 6,500 new MMLU-Pro and 1,000 new Omni-MATH problems, priced at the costs actually billed, predicted costs reduce spending at matched accuracy by 23.1% and 27.8% relative to median-length pricing. They also beat every alternative estimator we evaluated, including a reimplementation of ZeroRouter, with paired intervals excluding zero; the one exception is cost read off the success predictions on Omni-MATH, where length tracks difficulty. Deployed models drift: the same model served by different providers differs in output length and price. Treating each provider as the same model plus one offset, re-fitted from 50 calls, corrects a 37% overestimate after an unannounced provider shift, and routing over pinned providers saves 20% on APPS.

1. Introduction

Routing reasoning models requires predicting both correctness and cost before generation. Token rates are published; the number of tokens is not. A model that is cheap on average can be expensive on a particular problem, and the outputs of the models we study vary 10–90× across problems.

Prefill routers predict correctness from a small encoder’s prompt activations; the prefill router of Varshney et al. [12] explicitly prices outputs at median training length. We read output length from the same activations (Fig. 1). Cost prediction adds one linear readout per route to an existing prefill router, without another encoder pass.

We contribute: (i) evidence on fresh problems and billed prices that prefill cost readouts reduce spending at matched accuracy and outperform text-embedding, prompt-feature, difficulty-bin and success-derived cost estimators; (ii) a mechanism: success-derived pricing suffices where output length tracks difficulty and fails where it does not; (iii) a de-

ployment finding: the same model’s endpoints drift, and the shared readouts absorb this with one offset per endpoint.

Related work. MixLLM predicts output length from query embeddings [13]; CARROT estimates cost with embedding nearest neighbours or a fine-tuned RoBERTa [11]; ZeroRouter prices queries by item-response difficulty bins [16]. Activation-based length prediction also supports scheduling [10, 15]. Hosted open-weight models vary across providers in accuracy, length and price [2, 6, 9]; FACET routes among providers of a fixed model per (provider, task) using measured outcomes [3]. We instead route per query over endpoints of several models with predicted, endpoint-specific costs.

2. Cost Prediction and Routing

For query x and route m (a model at a fixed reasoning effort, served by an endpoint), let $p_m(x)$ be the probability of a correct answer and $c_m(x)$ the expected generation cost. One route is chosen before generation,

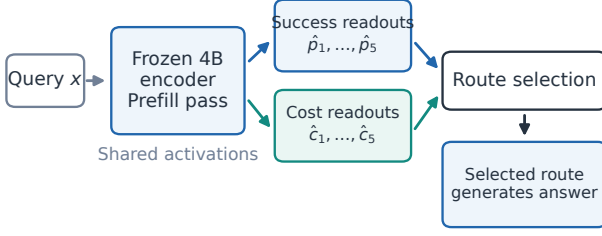
$$m^*(x; V) = \arg \max_m \{V \hat{p}_m(x) - \hat{c}_m(x)\}, \quad (1)$$

where V is the value of a correct answer; sweeping V trades accuracy for cost. Every query is answered once, with no verifier or fallback.

Readouts. We concatenate mean and last-token activations from eight layers of a frozen Qwen3-4B (Instruct-2507; Thinking-2507 on Omni), standardized on training data. Success readouts are L2-regularized logistic regressions with held-out Platt calibration; cost readouts are ridge regressions of log mean output length, regularized by training-only cross-validation. Predicted length is exponentiated with residual smearing and matched to the route’s training mean, then priced: $\hat{c}_m(x) = r_m^{\text{in}} n_m^{\text{in}}(x) + r_m^{\text{out}} \hat{\ell}_m(x)$.

Endpoint offsets. An endpoint of an already-modelled route (a new provider, or a provider whose behaviour changed) reuses the route’s readouts with one multiplicative length offset, the ratio of observed to predicted mean output over k labelled calls, and its own price.

(a) Routing architecture

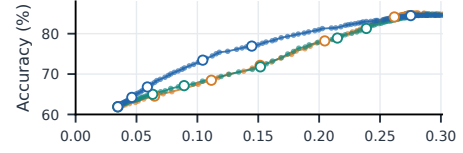


Route selection uses only the encoder activations.

● Training median length ● Training mean length ● Prefill cost prediction

Single-V policies; hollow circles mark calibration selections. Axes stop just past the first point within 0.5 pp of each arm's peak. Generation spending excludes encoder overhead.

(b) Fresh MMLU-Pro



Fresh Omni-MATH

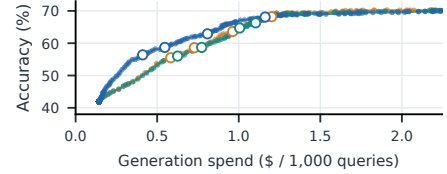


Figure 1. **Shared prefill readouts and fresh routing curves.** (a) One frozen encoder supplies per-route success and cost readouts. (b) Fresh accuracy versus spending for the deterministic policies of the V grid; circles mark policies selected on original calibration. Curves use list prices; Tables 1 and 2 use billed costs. Encoder overhead is excluded.

3. Experimental Protocol

The routes are gpt-oss-20b at low and medium effort, deepseek-v4-flash, and gpt-oss-120b at medium and high effort, called through OpenRouter. Original pools: 892 LiveCodeBench [5], 500 Omni-MATH-500 [1] and 1,000 subject-stratified MMLU-Pro [14] problems with 2–16 draws per route; train/calibration/test splits are 441/110/341, 275/75/150 and 550/150/300. Fresh sets: 6,500 MMLU-Pro and 1,000 Omni-MATH problems disjoint from the original pools (one draw per route), collected after all readouts were fitted.

Prices. Realized cost is the cost billed for each call. Predictions are priced at each model’s effective billed rate, fitted on the fresh calls (USD per million output tokens: 0.13 for gpt-oss-20b, 0.12 for deepseek-v4-flash, 0.26 for gpt-oss-120b). These differ from the list prices by up to $2.3\times$.

Evaluation. All estimators are fitted on original training problems only and evaluated on the fresh problems with the same success predictions. Each V on a dense grid gives a deterministic policy and an observed (cost, accuracy) point. Comparing two cost estimators a, b , we interpolate each one’s cost at 12 accuracies spanning the interior 90% of the range both reach and report $1 - \exp(\log C_a/C_b)$, the cost saved by a at matched accuracy. Intervals are percentile intervals from paired problem bootstraps (300 resamples), keeping each problem’s routes together. Separately, deployable policies select the cheapest V reaching each accuracy target on original calibration and are applied once to the fresh problems.

Cost saved by ours vs.	Omni-MATH ($n=1,000$)	MMLU-Pro ($n=6,500$)
Training-median length	27.8 [20.3, 33.1]	23.1 [18.1, 26.8]
Training-mean length	28.2 [22.4, 33.6]	20.5 [16.7, 23.6]
Prompt-feature GBM	20.3 [11.8, 26.5]	18.4 [12.2, 21.7]
ZeroRouter (reimplemented)	18.0 [9.0, 24.7]	19.3 [15.5, 22.6]
Difficulty bins	11.6 [5.2, 16.2]	12.7 [7.8, 16.1]
Cost from success predictions	3.2 [−1.7, 7.7]	14.1 [11.4, 16.6]

Table 1. **Fresh problems, billed costs.** Cost saved (%) by prefill cost readouts relative to each estimator at matched accuracy, with 95% paired bootstrap intervals. ZeroRouter: item-response latent fitted on the five routes, 4B features, its own success model and bin pricing, configuration chosen on calibration. Difficulty bins: ten quantile bins of mean predicted success logit.

Fresh set	Target	Savings, %	Δ acc., pp	Matched acc., %
MMLU-Pro	0.65	17.2 [9.8, 23.9]	+0.51 [−0.15, +1.18]	21.2 [11.7, 26.6]
	0.75	17.1 [10.2, 23.6]	+3.18 [+2.11, +4.19]	27.4 [19.5, 32.2]
	0.85	12.1 [9.0, 15.4]	−1.11 [−1.49, −0.69]	7.7 [3.8, 10.9]
Omni-MATH	0.65	15.3 [7.1, 22.9]	+1.80 [−0.10, +3.70]	20.9 [10.1, 28.0]
	0.70	16.8 [12.3, 21.4]	−1.70 [−3.20, −0.20]	7.5 [−3.1, 13.0]
	0.75	12.7 [8.4, 17.2]	−1.60 [−2.80, −0.40]	4.8 [−3.8, 9.0]

Table 2. **Deployable policies.** Policies chosen on original calibration for each accuracy target and applied once to fresh problems; learned versus median-length pricing, billed costs. Savings and Δ accuracy compare the two selected policies (paired bootstrap, 2,000); matched accuracy compares our policy with the median-pricing frontier at our achieved fresh accuracy (300 resamples). Fresh accuracy need not equal the target.

4. Results

Fresh problems. At matched accuracy, prefill cost readouts spend 23.1% less than median-length pricing on MMLU-Pro and 27.8% less on Omni-MATH (Table 1).

They also spend 18–20% less than a full ZeroRouter reimplementation and 18–20% less than a prompt-feature model. Calibration-selected policies save 12–29% at every target, but each lands at its own fresh accuracy, -1.7 to $+3.2$ points from the median policy (Table 2). Evaluated at our achieved accuracy, savings are 21–27% in the middle of the range and 5–8% near the top, where every method calls the strongest routes. Calibration selection itself costs little: our selected policies spend 0–2% more than our own fresh curve at the same accuracy, with one exception at 8%.

When success predictions suffice. Reading cost off the success predictions (a ridge regression of log length on success logits) ties the dedicated readout on Omni-MATH but loses 14.1 points on MMLU-Pro. On Omni-MATH, output length tracks difficulty. On MMLU-Pro, even empirical difficulty explains little of log length ($R^2 \approx .19$), while subject labels recover roughly 60% of the gap. The prefill encodes length information beyond difficulty, and that information pays where difficulty and length come apart.

Endpoints drift. On the fresh collection, the readouts overestimated deepseek-v4-flash’s cost by 37% (MMLU-Pro) and 23% (Omni-MATH); all other routes were within 8%. The aggregator had moved most of these calls to a provider absent from the original data, which writes about half as many tokens at equal accuracy. Re-fitting one length offset from 50 fresh calls restores calibration (predicted/realized 1.02 and 1.03). To test routing over providers, we pinned deepseek-v4-flash to three providers on 1,995 fresh MMLU-Pro and 957 APPS problems. Providers agree on correctness for 90–96% of problems, against 73–78% if independent, and their output lengths correlate at .83–.95: an endpoint is close to its model plus an offset. On APPS, where providers differ in accuracy (83–90%) and cost (up to $5\times$), routing over the three providers with offsets fitted on held-out problems saves 20.0% [10.7, 27.9] relative to the unpinned model and raises the best reachable accuracy from 85.3% to 89.5%. On MMLU-Pro, where providers agree, it saves 2.3% [−4.9, 7.7].

Original pools and contrasts. On the original test sets, savings against median pricing are 35.6% on LiveCodeBench, 21.9% on Omni-MATH and 30.7% on MMLU-Pro. On CodeContests [8], TACO [7], BigCodeBench [17] and APPS, oracle costs would save 20–36%, but predicted costs gain -3 to 14%, with intervals including zero. RouterBench [4] has 10.5% headroom. A fine-tuned Intern-Decision-4B success predictor, combined with our cost readouts, saves a further 11.2% [5.5, 17.1] on fresh Omni-MATH: improved success and cost predictors combine.

5. Discussion and Limitations

Each route needs output-length labels, and an endpoint offset needs a few labelled calls. All pools use five routes from two model families. Fresh sets have one draw per route, and APPS is the only coding pool with pinned providers. Intervals are pointwise, conditional on the fitted readouts, and assume independent problems. Billed rates were measured over one collection, and prices change within days. The ZeroRouter, MixLLM and CARROT baselines are reimplementations; the fresh CARROT and MixLLM comparisons are pending. The original-pool pipeline was corrected after outcomes were observed. Encoder overhead is excluded.

6. Conclusion

A frozen prefill that already predicts success also predicts cost. On fresh problems at billed prices, its cost readouts reduce spending at matched accuracy and outperform the alternative estimators we tested, most clearly where output length is not just difficulty. Because endpoints of one model share these readouts up to an offset, the router adapts to provider drift from a few dozen calls.

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